

Titanium Dioxide Nanoparticles in Food and Personal Care Products

Titanium dioxide is a common additive in many food, personal care, and other consumer products used by people, which after use can enter the sewage system, and subsequently enter the environment as treated effluent discharged to surface waters or biosolids applied to agricultural land, incinerated wastes, or landfill solids. This study quantifies the amount of titanium in common food products, derives estimates of human exposure to dietary (nano-) TiO₂, and discusses the impact of the nanoscale fraction of TiO₂ entering the environment. The foods with the highest content of TiO₂ included candies, sweets and chewing gums. Among personal care products, toothpastes and select sunscreens contained 1% to >10% titanium by weight. While some other crèmes contained titanium, despite being colored white, most shampoos, deodorants, and shaving creams contained the lowest levels of titanium (<0.01 µg/mg). For several high-consumption pharmaceuticals, the titanium content ranged from below the instrument detection limit (0.0001 µg Ti/mg) to a high of 0.014 µg Ti/mg. Electron microscopy and stability testing of food-grade TiO₂ (E171) suggests that approximately 36% of the particles are less than 100 nm in at least one dimension and that it readily disperses in water as fairly stable colloids. However, filtration of water solubilized consumer products and personal care products indicated that less than 5% of the titanium was able to pass through 0.45 or 0.7 µm pores. Two white paints contained 110 µg Ti/mg while three sealants (i.e., prime coat paint) contained less titanium (25 to 40 µg Ti/mg). This research showed that while many white-colored products contained titanium, it was not a prerequisite. Although several of these product classes contained low amounts of titanium, their widespread use and disposal down the drain and eventually to WWTPs deserves attention. A Monte Carlo human exposure analysis to TiO₂ through foods identified children as having the highest exposures because TiO₂ content of sweets is higher than other food products, and that a typical exposure for a US adult may be on the order of 1 mg Ti per kilogram body weight per day. Thus, because of the millions of tons of titanium based white pigment used annually, testing should focus on food-grade TiO₂ (E171) rather than that adopted in many environmental health and safety tests (i.e., P25), which is used in much lower amounts in products less likely to enter the environment (e.g., catalyst supports, photocatalytic coatings).